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-- =====
-- BLOOD DONATION MANAGEMENT SYSTEM (BDMS) - Sargodha, Pakistan
-- MySQL (XAMPP) + Python (Flask) + HTML
-- THREE ROLES: Admin, Donor, Recipient
-- =====

DROP DATABASE IF EXISTS bdms;
CREATE DATABASE bdms CHARACTER SET utf8mb4 COLLATE utf8mb4_general_ci;
USE bdms;

CREATE TABLE blood_bank (
    bank_id INT AUTO_INCREMENT PRIMARY KEY,
    bank_name VARCHAR(100) NOT NULL,
    address VARCHAR(150) NOT NULL,
    city VARCHAR(50) NOT NULL DEFAULT 'Sargodha',
    contact VARCHAR(20) NOT NULL
);

CREATE TABLE hospital (
    hospital_id INT AUTO_INCREMENT PRIMARY KEY,
    hospital_name VARCHAR(100) NOT NULL,
    address VARCHAR(150) NOT NULL,
    city VARCHAR(50) NOT NULL DEFAULT 'Sargodha',
    contact VARCHAR(20) NOT NULL
);

-- DONOR entity
CREATE TABLE donor (
    donor_id INT AUTO_INCREMENT PRIMARY KEY,
    full_name VARCHAR(100) NOT NULL,
    cnic VARCHAR(15) NOT NULL UNIQUE,
    gender ENUM('Male','Female') NOT NULL,
    dob DATE NOT NULL,
    blood_group ENUM('A+','A-','B+','B-','AB+','AB-','O+','O-') NOT NULL,
    phone VARCHAR(15) NOT NULL,
    email VARCHAR(100),
    address VARCHAR(150) NOT NULL,
    city VARCHAR(50) NOT NULL DEFAULT 'Sargodha',
    last_donation_date DATE,
    registered_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- RECIPIENT entity (separate from donor, as in the ER diagram)
CREATE TABLE recipient (

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recipient_id INT AUTO_INCREMENT PRIMARY KEY,
full_name   VARCHAR(100) NOT NULL,
cnic        VARCHAR(15) NOT NULL UNIQUE,
gender      ENUM('Male','Female') NOT NULL,
dob         DATE NOT NULL,
blood_group ENUM('A+','A-','B+','B-','AB+','AB-','O+','O-') NOT NULL,
phone       VARCHAR(15) NOT NULL,
email       VARCHAR(100),
address     VARCHAR(150) NOT NULL,
city        VARCHAR(50) NOT NULL DEFAULT 'Sargodha',
registered_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- USERS login accounts (Chapter 11 security). role = admin / donor / recipient
-- A donor login links to a donor row; a recipient login links to a recipient row.
CREATE TABLE users (
  user_id   INT AUTO_INCREMENT PRIMARY KEY,
  username  VARCHAR(50) NOT NULL UNIQUE,
  password  VARCHAR(255) NOT NULL,
  role      ENUM('admin','donor','recipient') NOT NULL DEFAULT 'donor',
  donor_id  INT,
  recipient_id INT,
  created_at  TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (donor_id) REFERENCES donor(donor_id) ON DELETE SET NULL,
  FOREIGN KEY (recipient_id) REFERENCES recipient(recipient_id) ON DELETE SET NULL
);

CREATE TABLE blood_unit (
  unit_id   INT AUTO_INCREMENT PRIMARY KEY,
  blood_group ENUM('A+','A-','B+','B-','AB+','AB-','O+','O-') NOT NULL,
  bank_id   INT NOT NULL,
  donor_id  INT,
  status    ENUM('available','reserved','used') NOT NULL DEFAULT 'available',
  collected_date DATE NOT NULL,
  FOREIGN KEY (bank_id) REFERENCES blood_bank(bank_id),
  FOREIGN KEY (donor_id) REFERENCES donor(donor_id) ON DELETE SET NULL
);

CREATE TABLE appointment (
  appointment_id INT AUTO_INCREMENT PRIMARY KEY,
  donor_id       INT NOT NULL,
  bank_id        INT NOT NULL,
  appointment_date DATE NOT NULL,
  status         ENUM('scheduled','completed','cancelled') NOT NULL DEFAULT 'scheduled',

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FOREIGN KEY (donor_id) REFERENCES donor(donor_id) ON DELETE CASCADE,
FOREIGN KEY (bank_id) REFERENCES blood_bank(bank_id)
);

-- BLOOD REQUEST now links to a RECIPIENT (and a hospital)
CREATE TABLE blood_request (
    request_id INT AUTO_INCREMENT PRIMARY KEY,
    recipient_id INT,
    hospital_id INT NOT NULL,
    patient_name VARCHAR(100) NOT NULL,
    blood_group ENUM('A+', 'A-', 'B+', 'B-', 'AB+', 'AB-', 'O+', 'O-') NOT NULL,
    units_required INT NOT NULL DEFAULT 1,
    status ENUM('pending', 'approved', 'rejected', 'fulfilled') NOT NULL DEFAULT 'pending',
    request_date DATE NOT NULL,
    FOREIGN KEY (recipient_id) REFERENCES recipient(recipient_id) ON DELETE SET NULL,
    FOREIGN KEY (hospital_id) REFERENCES hospital(hospital_id)
);

CREATE TABLE donation_history (
    donation_id INT AUTO_INCREMENT PRIMARY KEY,
    donor_id INT NOT NULL,
    bank_id INT NOT NULL,
    donation_date DATE NOT NULL,
    blood_group ENUM('A+', 'A-', 'B+', 'B-', 'AB+', 'AB-', 'O+', 'O-') NOT NULL,
    units INT NOT NULL DEFAULT 1,
    FOREIGN KEY (donor_id) REFERENCES donor(donor_id) ON DELETE CASCADE,
    FOREIGN KEY (bank_id) REFERENCES blood_bank(bank_id)
);

-- Indexes (Chapter 5 - physical design)
CREATE INDEX idx_unit_group ON blood_unit(blood_group);
CREATE INDEX idx_unit_status ON blood_unit(status);
CREATE INDEX idx_donor_group ON donor(blood_group);
CREATE INDEX idx_recipient_group ON recipient(blood_group);
CREATE INDEX idx_request_status ON blood_request(status);

-- ===== SAMPLE DATA (Sargodha) =====

INSERT INTO blood_bank (bank_name, address, city, contact) VALUES
('DHQ Hospital Blood Bank', 'Civil Lines, Sargodha', 'Sargodha', '048-9230301'),
('Sargodha Red Crescent Blood Bank', 'Club Road, Sargodha', 'Sargodha', '048-3768900'),
('Niazi Welfare Blood Bank', 'University Road, Sargodha', 'Sargodha', '048-3220011');

INSERT INTO hospital (hospital_name, address, city, contact) VALUES

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('DHQ Teaching Hospital Sargodha', 'Civil Lines, Sargodha', 'Sargodha', '048-9230300'),
('Niazi Medical & Dental Hospital', 'Faisalabad Road, Sargodha', 'Sargodha', '048-3768111'),
('Ittefaq Hospital Sargodha', 'Khushab Road, Sargodha', 'Sargodha', '048-3740555'),
('Mansoor Hospital', 'Satellite Town, Sargodha', 'Sargodha', '048-3211234');
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INSERT INTO donor (full_name, cnic, gender, dob, blood_group, phone, email, address, city,
last_donation_date) VALUES
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```
('Muhammad Ahmed', '38401-1234567-1', 'Male', '1995-03-12', 'O+', '0300-1234567',
'ahmed@gmail.com', 'Satellite Town, Sargodha', 'Sargodha', '2026-02-10'),
('Fatima Bibi', '38401-2345678-2', 'Female', '1998-07-25', 'B+', '0321-2345678',
'fatima@gmail.com', 'University Road, Sargodha', 'Sargodha', '2026-01-15'),
('Bilal Hassan', '38401-3456789-3', 'Male', '1992-11-05', 'A+', '0333-3456789',
'bilal@gmail.com', 'Khushab Road, Sargodha', 'Sargodha', '2026-03-01'),
('Ayesha Khan', '38401-4567890-4', 'Female', '2000-01-30', 'AB+', '0345-4567890',
'ayesha@gmail.com', 'Civil Lines, Sargodha', 'Sargodha', NULL),
('Usman Ali', '38401-5678901-5', 'Male', '1990-06-18', 'O-', '0301-5678901',
'usman@gmail.com', 'Club Road, Sargodha', 'Sargodha', '2025-12-20'),
('Zainab Tariq', '38401-6789012-6', 'Female', '1997-09-09', 'A-', '0322-6789012',
'zainab@gmail.com', 'Jail Road, Sargodha', 'Sargodha', NULL),
('Hamza Sheikh', '38401-7890123-7', 'Male', '1994-04-22', 'B-', '0334-7890123',
'hamza@gmail.com', 'Block 7, Sargodha', 'Sargodha', '2026-02-28'),
('Mehwish Akram', '38401-8901234-8', 'Female', '1999-12-14', 'O+', '0346-8901234',
'mehwish@gmail.com', 'Faisalabad Road, Sargodha', 'Sargodha', NULL);
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INSERT INTO recipient (full_name, cnic, gender, dob, blood_group, phone, email, address, city)
VALUES
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```
('Imran Yousaf', '38401-1112223-1', 'Male', '1988-05-10', 'O+', '0300-1112223',
'imran@gmail.com', 'Satellite Town, Sargodha', 'Sargodha'),
('Saima Riaz', '38401-2223334-2', 'Female', '1993-08-19', 'A+', '0321-2223334',
'saima@gmail.com', 'Civil Lines, Sargodha', 'Sargodha'),
('Kashif Mehmood', '38401-3334445-3', 'Male', '1985-02-27', 'B+', '0333-3334445',
'kashif@gmail.com', 'Khushab Road, Sargodha', 'Sargodha'),
('Nadia Aslam', '38401-4445556-4', 'Female', '1996-11-03', 'AB+', '0345-4445556',
'nadia@gmail.com', 'University Road, Sargodha', 'Sargodha');
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-- Login accounts (passwords HASHED with pbkdf2:sha256)
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-- admin / admin123 (admin)
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-- ahmed / ahmed123 (donor) -> donor 1
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-- fatima / fatima123 (donor) -> donor 2
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-- imran / imran123 (recipient) -> recipient 1
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-- saima / saima123 (recipient) -> recipient 2
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INSERT INTO users (username, password, role, donor_id, recipient_id) VALUES
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('admin',
'pbkdf2:sha256:1000000$8g5VJqxXxgXHMYkm$005f0a9bbbeb532c019687ce9f3a3074f682c29
cbe101d69a54ce36815f1e7d6', 'admin', NULL, NULL),
('ahmed',
'pbkdf2:sha256:1000000$ojWBt1R56umJlrM6$29710009bfbfc5736b26f9e43421c0f56cb1e3b6c
72b3906ef79aff7efbd817f', 'donor', 1, NULL),
('fatima',
'pbkdf2:sha256:1000000$jBp4bCFAsO83kWp7$682bd7a1afbfb946184a5fe1f933cd647a16fe39
3a26562a8b2e3eff2582411e', 'donor', 2, NULL),
('imran',
'pbkdf2:sha256:1000000$AcraNCwmMR0clFdW$3f5e29f81af33527a3412336bd940869ca3940
af0232e655a0849af2f039ca19', 'recipient', NULL, 1),
('saima',
'pbkdf2:sha256:1000000$3YKfFJHeLzAa8eFN$67a873492de9194f2e338ef178d7abfb74c7f0d0
76ae79e9eae40f370c761f56', 'recipient', NULL, 2);

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INSERT INTO blood_unit (blood_group, bank_id, donor_id, status, collected_date) VALUES
('O+', 1, 1, 'available', '2026-02-10'),
('O+', 1, 8, 'available', '2026-06-01'),
('B+', 2, 2, 'available', '2026-01-15'),
('A+', 1, 3, 'available', '2026-03-01'),
('A+', 2, NULL, 'available', '2026-05-20'),
('O-', 3, 5, 'used', '2025-12-20'),
('B-', 1, 7, 'available', '2026-02-28'),
('AB+', 2, NULL, 'available', '2026-04-11'),
('O+', 3, NULL, 'reserved', '2026-05-30');

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INSERT INTO appointment (donor_id, bank_id, appointment_date, status) VALUES
(4, 1, '2026-06-25', 'scheduled'),
(6, 2, '2026-06-28', 'scheduled'),
(1, 1, '2026-02-10', 'completed');

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INSERT INTO blood_request (recipient_id, hospital_id, patient_name, blood_group,
units_required, status, request_date) VALUES
(1, 1, 'Imran Yousaf', 'O+', 2, 'pending', '2026-06-18'),
(2, 2, 'Saima Riaz', 'A+', 1, 'approved', '2026-06-17'),
(3, 3, 'Kashif Mehmood', 'B+', 1, 'fulfilled', '2026-06-10');

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INSERT INTO donation_history (donor_id, bank_id, donation_date, blood_group, units)
VALUES
(1, 1, '2026-02-10', 'O+', 1),
(2, 2, '2026-01-15', 'B+', 1),
(3, 1, '2026-03-01', 'A+', 1),
(5, 3, '2025-12-20', 'O-', 1),

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(7, 1, '2026-02-28', 'B-', 1);